

9/30/2026

We have proved:

Thm (Unique factorization in $\mathbb{Z}[\sqrt{d}]$, $d = -1, \pm 2, 3$)

Every nonzero, nonunit element of $\mathbb{Z}[\sqrt{d}]$ can be expressed as a product of primes, uniquely up to order and multiplication by units.

Q: What are units and primes in $\mathbb{Z}[\sqrt{d}]$?

We will start with $d < 0$.

Units

$\alpha \in \mathbb{Z}[\sqrt{d}]$ is a unit $\Leftrightarrow |N(\alpha)| = 1$.

Write $\alpha = x + y\sqrt{d}$. Then

$$|N(\alpha)| = |x^2 - dy^2| = x^2 - dy^2.$$

So

$$\alpha \text{ is a unit} \Leftrightarrow x^2 - dy^2 = 1.$$

$\uparrow$

Diophantine equation

Case I: $d = -1$.

$$x^2 + y^2 = 1 \Rightarrow (x, y) = (\pm 1, 0), (0, \pm 1)$$

Remember $\alpha = x + yi$. So $\alpha = \pm 1, \pm i$.

Case II. $d < -1$.

$$x^2 - \underbrace{dy^2}_{>1} = 1 \Rightarrow y = 0, x = \pm 1.$$

Recall $\alpha = x + y\sqrt{d}$. So $\alpha = \pm 1$

Ex. Consider the case $d = -m^2$. Show that $\mathbb{Z}[\sqrt{d}] =$

$\mathbb{Z}[m_i]$ is not a UFD by looking at

$$-d = m^2 = m \cdot m = (-1)m_i \cdot m_i$$

and showing that m_i is prime.

Primes

Consider Gaussian integers $\mathbb{Z}[i]$ ($d = -1$).

$\forall \alpha \in \mathbb{Z}[i]$, say $\alpha = x + iy$, then it is prime if

$$N(\alpha) = x^2 + y^2 = p \text{ is a prime number.}$$

We know that this can happen exactly when $p = 2$ or

$$p \equiv 1 \pmod{4}.$$

What is the "smallest" prime(s) in $\mathbb{Z}[i]$?

These are elements $\alpha = x + yi$ with

$$N(\alpha) = x^2 + y^2 = 2.$$

So $(x, y) = (\pm 1, \pm 1)$. Hence $\alpha = \pm 1 \pm i$, namely,

$1+i$ and its associates $-(1+i)$, $i(1+i)$, $-i(1+i)$.

When $p \equiv 1 \pmod{4}$, we know there are x, y s.t.

$$N(\alpha) = \alpha \bar{\alpha} = x^2 + y^2 = p.$$

So p is not prime in $\mathbb{Z}[i]$.

If $p \equiv 3 \pmod{4}$, is it prime in $\mathbb{Z}[i]$?

Let $\beta \mid p$. Then $N(\beta) \mid N(p) = p^2$. So

$$N(\beta) = 1, p, p^2.$$

$\downarrow$
units

$\downarrow$
associates of p .

If $N(\beta) = p$ and $\beta = x + yi$, then

$$N(\beta) = x^2 + y^2 = p,$$

impossible because $p \equiv 3 \pmod{4}$. So p is prime in $\mathbb{Z}[i]$.

Lem. Every prime $\pi \in \mathbb{Z}[i]$ divides exactly one prime number p .

Pf. $\pi \mid N(\pi) = \pi \bar{\pi}$. Factor

$$N(\pi) = p_1^{e_1} p_2^{e_2} \dots p_k^{e_k},$$

where $p_1, \dots, p_k$ are distinct. Euclid's lemma for $\mathbb{Z}[i]$

implies $\pi \mid p_i$ for some $1 \leq i \leq k$.

If $\pi \mid p$ and $\pi \mid q$, where $p \neq q$, then $(p, q) = 1$

in $\mathbb{Z}$. Bezout's lemma for $\mathbb{Z}$ gives $x, y \in \mathbb{Z}$ s.t.

$$px + qy = 1.$$

$\pi \mid p$, $\pi \mid q \Rightarrow \pi \mid px + qy = 1 \Rightarrow \pi$ is a unit. X

So π can divide only one prime number.

